الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني

الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام 2021 الثانية

الاسم والشهرة: ----------

رقم المرشح: ----------

الاختصاص والرمز : الالكترونيك (20203)
المادة : الالكترونيك الرقمي
المستندات المسموح بها : آلة حاسبة غير مبرمجة

المعدل : (10)
المدة : ساعتان

الإجابات على "كراس الإجابة"

Part A- Choose the correct answer: (1 point/ correct answer)

1- If register TRISA = $(0F)_H$, then PORTA will be initiated as :
a) Input
b) Half output – half input
c) Output
d) Nothing

2- If $(0x21) = (53)_H$ and we execute the instruction [CLRF 0x21], the result will be :
a) $W = (00)_H$
b) $W = (53)_H$
c) $(0x21) = (53)_H$
d) $(0x21) = (00)_H$

3- The result of the instruction [ADDLW 0x01] with $W = (FF)_H$ will be :
a) $W = (00)_H$
b) $W = (01)_H$
c) $W = (FF)_H$
d) $W = (EF)_H$

4- To move to page 0 (BANK0), we write :
a) BCF STATUS,RP0
b) BSF STATUS,RP0
c) BCF TRISB,0
d) BSF TRISB,0

5- To turn ON a LED which anode is connected to pin RA2 and cathode to the ground, we erite :
a) BSF PORTA, 2
b) BCF PORTA, 2
c) BSF TRISA, 2
d) BCF TRISA, 2

6- If DDRB = $00, so PORTB of the PIA will be initiated as :
a) Output
b) Half input– half output
c) Input
d) Nothing

7- If DDRA = $F0 so PORTA of the PIA 6821 will be initiated as :
a) Half input– half output
b) All output
c) All input
d) Nothing

8- The result of the instruction [ORA #$AA] with A = $FF will be :
a) A = $FF
b) A = $FA
c) A = $AA
d) A = $00

9- The result of the instruction [EORA #$A0] with A = $FF will be :
a) A = $FF
b) A = $FA
c) A = $AA
d) A = $5F

10- To turn ON a LED which cathode is connected to pin PA3 and anode to Vcc , we write CLRB and:
a) STA PORTA
b) STA ORB
c) STB ORA
d) STB ORB

TS 2021 Micro-2nd

Part A.

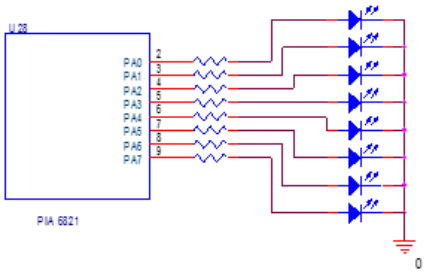
	Ans.	Discussion.
1)	b.	TRISA = {0 = output, 1 = Input} ⇒ TRISA = $0F half output half input
2)	d.	Clf 0x21 will make (0x21) = $(60)_H$.
3)	a.	W. = $(FF)_H$ FF +01 100 W ←
4).	a.	Clear RP0 will select Bank0 BCF STATUS, RP0
5)	a.	set PORTA,2 will make the led ON. RA2 — Anode — Cathod
6)	C.	DDRB = {00 = Input, FF = output.}
7).	a.	DDRA = {00 = Input, FF = output} ⇒ DPRA = $F0 half input half output
8)	a).	A = $FF → ORA #$AA ?? FF = 11111111 AA = 10101010 OR A ← (1111 1111)
9)	d.	EOR A ≡ XOR. FF = 11111111 A0 = 10100000 0101 1111 A ← (5 F) XOR: A B Z 0 0 0 0 1 1 1 0 1 1 1 0
10)	C.	Clear Acc.B Store B in portA RA3. cathod — Anode — +Vcc

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Part B: Treat 2 questions only : (5 pts / question)

I- Write the program that allows to sequentially displaying at port A of a PIA 6821 of MP 6809 the bytes in the table below, with a delay (of your choice) between two consecutive displays.

PA7	PA6	PA5	PA4	PA3	PA2	PA1	PA0
1	1	0	0	0	0	1	1
0	1	1	0	0	1	1	0
0	0	0	1	1	0	0	0
0	0	1	0	0	1	0	0
0	1	0	0	0	0	1	0
1	0	0	0	0	0	0	1
1	1	0	0	0	0	1	1
1	1	1	1	1	1	1	1



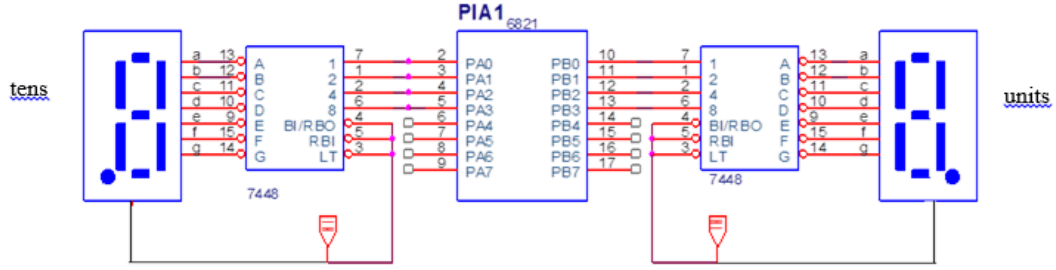
1)

```
DDRA EQU $8000
PORTA EQU $8000
CRA  EQU $8001
ORG $FC00
CLR CRA
CLR DDRA
COM DDRA
LDA #$04
STA CRA
START: LDA #$C0
STA $1000
LDA #$03
STA $1001
```

```
Loop1: LDA $1000
ORA $1001
Delay JSR Delay.
STA PORTA
LSL $1001
LSR $1000
BCC Loop1
STA PORTA
JSR Delay
LDA #$C3
STA PORTA
JSR Delay.
LDA #$FF
STA PORTA
BRA START
Delay:
CLRB
Loop2:
INCB
BCC Loop2
RTS
END
```

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II- By referring to the below circuit, write the program to build a **decimal down** ***counter*** from ***88*** to ***00***.

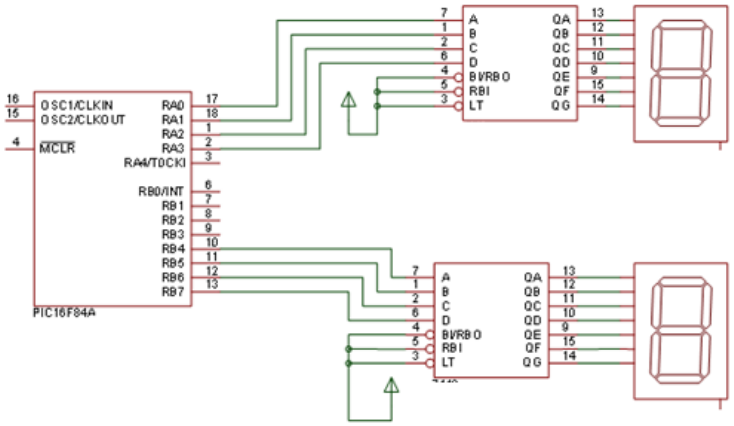


2).

```
DDRA EQU $8000
PORTA EQU $8000
CRA EQU $8001
DDRB EQU $8002
PORTB EQU $8002
CRA EQU $8003
ORG $FC00
CLR CRA
CLR CRB
CLR DDRA
COM DDRA
CLR DDRB
COM DDRB
LDA #$01
STA CRA
STA CRB
START:
LDA #$08
STA PORTA
STA PORTB
LOOP1: DEC PORTB
BNE LOOP1
LDA #$09
STA PORTB
DEC PORTA
BNE LOOP1
BRA START
END
```

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III- By referring to the below circuit, write the program to build a **decimal counter from 00 to 98**. PORTA is for tens and PORTB is for units. Use a delay of your choice between two consecutive displays.



3) #include (Pic16f84a.inc)

```
count1 equ 0x20
count2 equ 0x21
org 0x00
bsf STATUS,RP0
clrf TRISA
clrf TRISB
bcf STATUS,RP0
START
  clrf count1
  clrf count2
  clrf PORTA
  clrf PORTB
```

```
main
  Call Delay
  swap Count1,W
  MOVWF PORTB
  incf count1
  BTFSC Count1,3
  BTFSS Count1,1
  goto main
  Clrf Count1.
  Incf Count2
  movF Count2,W
  MOVWf PORTA
  BTFSC Count2,3
  BTFSS Count2,1
   goto main
   goto START
   end

  Delay
   clrf 0x22
   loop
   Incfsz 0x22
   goto loop
   Return
```

ANNEXE / APPENDIX

Instruction set of 6809

ADCA, ADCB, ADDA, ADDB, ADDD, ANDA, ANDB, ANDCC, ASLA, ASLB, ASL, ASRA, ASRB, ASR, BITA, BITB, CLRA, CLRB, CLR, CMPA, CMPB, CMPD, CMPU, CMPS, CMPX, CMPY, COMA, COMB, COM, CWAI, DAA, DECA, DECB, DEC, EORA, EORB, EXG, INCA, INCB, INC, JMP, JSR, LDA, LDB, LDD, LDS, LDU, LDX, LDY, LEAS, LEAU, LEAX, LEAY, LSLA, LSLB, LSL, LSRA, LSRB, LSR, MUL, NEGA, NEGB, NEG, NOP, ORA, ORB, ORCC, PSHS, PSHU, PULS, PULU, ROLA, ROLB, ROL, RORA, RORB, ROR, RTI, RTS, SBCA, SBCB, SEX, STA, STB, STD, STS, STU, STX, STY, SUBA, SUBB, SUBD, SWI, SWI2, SWI3, SYNC, TFR, TSTA, TSTB, TST.

PIA:

ORA et DDRA = $8000	**CRA = $8001**
ORB et DDRB = $8002	**CRB = $8003**

Instruction set of PIC 16F84A

ADDWF, ANDWF, CLRF, CLRW, COMF, DECF, DECFSZ, INCF, INCFSZ, IORWF, MOVF, MOVWF, NOP, RLF, RRF, SUDWF, SWAPF, XORWF, BCF, BSF, BTFSC, BTFSS, ADDLW, ANDLW, CALL, CLRWDT, GOTO, IORLW, MOVLW, RETFIE, RETLW, RETURN, SLEEP, SUBLW, XORLW.

Internal architecture of PIC16F84A

File Address			File Address
00h	Indirect addr.(1)	Indirect addr.(1)	80h
01h	TMR0	OPTION_REG	81h
02h	PCL	PCL	82h
03h	STATUS	STATUS	83h
04h	FSR	FSR	84h
05h	PORTA	TRISA	85h
06h	PORTB	TRISB	86h
07h	—	—	87h
08h	EEDATA	EECON1	88h
09h	EEADR	EECON2(1)	89h
0Ah	PCLATH	PCLATH	8Ah
0Bh	INTCON	INTCON	8Bh
0Ch	General Purpose Registers (SRAM)	Mapped (accesses) in Bank 0	8Ch
			FFh
	Bank 0	Bank 1	

PS2:PS0: Prescaler Rate Select bits

Bit Value	TMR0 Rate	WDT Rate
000	1 : 2	1 : 1
001	1 : 4	1 : 2
010	1 : 8	1 : 4
011	1 : 16	1 : 8
100	1 : 32	1 : 16
101	1 : 64	1 : 32
110	1 : 128	1 : 64
111	1 : 256	1 : 128